

Identifying the key obstacle in photocatalytic oxygen evolution on rutile TiO₂

Dong Wang^{1,3}, Tian Sheng^{1,3}, Jianfu Chen¹, Hai-Feng Wang^{1*} and P. Hu^{1,2*}

As the bottleneck in photocatalytic water splitting, the oxygen evolution reaction (OER) has drawn huge attention, but its efficiency still falls short of expectations. A widely accepted speculation is that the catalysts' activity is insufficient (high reaction barriers need to be overcome). Here, we develop a first-principles method to investigate the photocatalytic OER at the water/TiO₂(110) interface. A full mechanism uncovering the importance of radicals is determined. Kinetic analysis further enables to quantitatively estimate each possible obstacle in the process. We demonstrate unambiguously that the rate-determining factor of the OER varies with the concentration of surface-reaching photoholes (C_{h+}). Under experimental conditions, the intrinsic catalytic activity of TiO₂(110) does not represent the main obstacle, but all steps involving the photoholes are slow due to their low concentrations. This suggests that the key to enhance the OER efficiency is to increase C_{h+} before C_{h+} reaches the estimated threshold (C_{h+} = ~10⁻⁴).

As an environmentally friendly approach to generate renewable energy directly from the Sun, potentially contributing to sustainable energy development, photocatalytic water splitting ($\text{H}_2\text{O} \xrightarrow{h\nu} \frac{1}{2}\text{O}_2 + \text{H}_2$) has drawn huge attention in chemistry^{1,2}. Overall, this process consists of the hydrogen evolution reaction ($\text{H}^+ + \text{e}^- \rightarrow \frac{1}{2}\text{H}_2$) and the OER involving photoholes h^+ ($2\text{H}_2\text{O} + 4h^+ \rightarrow \text{O}_2 + 4\text{H}^+$) (Fig. 1a), where the OER is known to be the bottleneck, hindering the overall process³. Therefore, huge efforts have been made to increase the efficiency of the OER on a variety of catalysts, especially on TiO₂, which is perhaps one of the most important materials in photocatalysis due to its natural abundance, low cost, non-toxicity and superior photostability^{2,3}.

Among the different extensively studied topics, the reaction mechanism of the OER has been the main focus both experimentally and theoretically. It is anticipated that further insight into the mechanism can provide guidance to rationally improve the catalytic efficiency. However, previous studies have fallen short on their promises due to the complicated nature of the OER and the limitations of the approaches used. Based on the experimental work, mainly two mechanisms were suggested. It has long been speculated that O₂ is produced from H₂O₂ by coupling two surface ·OH radicals⁴. But later spectroscopy results excluded the generation of H₂O₂ in the OER^{5,6}. Subsequently, a nucleophilic attack mechanism was proposed, suggesting that the O–O bond is formed by involvement of lattice O (refs 5,7,8). Theoretically, few studies were reported on the overall OER at the liquid/solid interface due to the difficulties in simulating surface radicals (trapped holes on surface) and the aqueous environment. Most works were limited on the initial water dissociation step^{9,10} with few exploratory studies focused on the thermodynamics of the process^{11–13}. It is clear that due to the limitations in both experimental and theoretical approaches, the mechanism of the OER remains elusive.

In addition to the mechanism, extensive investigations have also been carried out on a variety of other issues, such as the photocatalyst preparation^{1,2}, the band structure modification (for example, ion doping^{14–16}), increasing the efficiency of photoinduced charge

separation (for example, phase junction^{17–19}) and lowering the reaction barriers (for example, the water dissociation barrier^{20,21}). It is clear that there is still a lack of understanding on the main obstacle in the OER and therefore it is not surprising that the overall efficiency of the OER is still unsatisfactory. Obviously, several fundamental questions need to be answered to achieve a breakthrough. First, what is the favoured reaction mechanism of the photocatalytic OER? Second, which factor determines the overall efficiency? Third, how can we further improve the efficiency?

In this work, we chose the most stable rutile TiO₂(110) surface as the model photocatalyst (Fig. 1a), considering that the rutile phase is generally good for water oxidation^{22,23} compared with the anatase phase^{24,25}. A systematic computational investigation into the photocatalytic OER mechanism at the water/TiO₂(110) interface was carried out, using extensive first-principles molecular dynamics (MD) simulations. A comprehensive picture of the whole chemical process, consisting of H₂O dissociation, the formation of active radical species and the O–O coupling reaction, is obtained. Based on the completed reaction mechanism and detailed microkinetic analyses, we demonstrate that under experimental conditions the intrinsic catalytic activity of TiO₂ is not the rate-determining factor that limits the overall efficiency of the OER. It is shown clearly that the low concentration of surface-reaching holes is the main obstacle in the system.

Results

Development of the multi-point averaging molecular dynamics (MPA-MD) approach. Theoretically, it is extremely difficult to calculate the OER since one has to overcome two main issues. The first is to correctly locate the photoinduced surface radicals, which is a prerequisite to test a variety of pathways in the OER. As pointed out in ref. 26, the error of the commonly used PBE functional for calculating radicals can be as large as 0.6 eV, but the problem can be solved by using hybrid functional calculations, such as HSE06²⁶. However, the HSE06 functional is too time consuming for the OER system, particularly in the presence of the liquid phase. In our

¹Key Laboratory for Advanced Materials, Center for Computational Chemistry and Research Institute of Industrial Catalysis, School of Chemistry and Molecular Engineering, East China University of Science and Technology, Shanghai, China. ²School of Chemistry and Chemical Engineering, Queen's University Belfast, Belfast, UK. ³These authors contributed equally: Dong Wang and Tian Sheng. *e-mail: hfwang@ecust.edu.cn; p.hu@qub.ac.uk

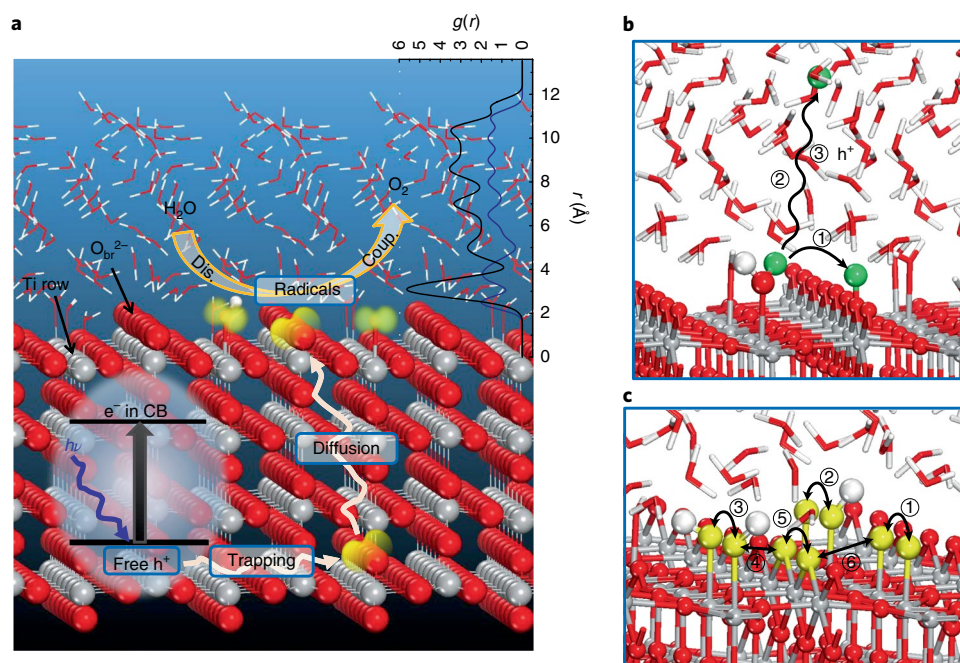


Fig. 1 | Schematic of the photocatalytic OER at the water/TiO₂ interface. a, The general picture of the OER. A four-Ti-layer $p(1 \times 4)$ periodical slab with 26 H₂O above the surface is used. Ti sites and bridge O are labelled and the calculated densities $g(r)$ of O and H varying with the distance $r(\text{\AA})$ above the TiO₂ surface in the water phase are shown by the blue and black curves, respectively. Dis., dissociation; Coup., coupling. **b,c**, Possible pathways of water dissociation (**b**) and O–O coupling (**c**) among surface radicals (highlighted in yellow): (1) coupling between O_i[•] and O_i[•]; (2) $\cdot\text{OH}_i$ and O_i[•]; (3) $\cdot\text{OH}_i$ and $\cdot\text{OH}_j$; (4) $\cdot\text{OH}_i$ and O_{br}[•]; (5) O_{br}[•] and O_{br}[•]; (6) O_i[•] and O_{br}[•]. Grey, Ti; red, O; white, H; green, the reacting H in water dissociation. This colour notation is used throughout the paper.

previous work^{27,28}, we showed that DFT+U can yield similar structures as the HSE06 functional with reasonable energies (Supplementary Fig. 1 and Supplementary Note 1). In this work, we first used the DFT+U method to run MD simulations as well as the structure optimization of each MD snapshot, and then all the selected samples from each MD simulation were further optimized/checked by the HSE06 functional. On using this technique, almost insurmountable hybrid functional calculations in this project could be carried out within a reasonable timescale; without this approach, it is perhaps impossible to test a variety of OER pathways as we did in the current work.

The second issue is to reliably calculate the reactions occurring at the liquid/solid interface. The explicit involvement of the water environment is known to be important to the reaction energetics^{29,30}, which is also a prerequisite to investigate the proton transfer process commonly occurring at the liquid/solid interface. But there is no robust method for such systems using density functional theory (DFT) approaches. To calculate the OER reactions with good accuracy at the water/TiO₂ interface (Fig. 1a), we developed a general method, the so-called MPA-MD approach. First, we performed the normal MD calculations (~9 ps; see Methods) of each intermediate state (IMS; including transition states) on TiO₂(110) with the aqueous network above the surface. Second, in the MD simulation of each IMS, we selected a structure every 0.2 ps from the stabilized MD simulations (small fluctuation in energy after long simulations) and further optimized it to obtain the total energy of each structure (E_{tot}) at the HSE06 level. For each IMS, around 15 samples from the late part of each overall MD simulation (~3 ps) were obtained. Considering that different solution configurations of the water network affect E_{tot} , we deducted the contribution of the water solution in E_{tot} but still considered the solvation effect for each sample as follows: (i) calculate the total energy of the water solution (E_{water}) with exactly the same structure as in the optimized samples at the HSE06 level; (ii) deduct E_{water} from E_{tot} to obtain the solvation-included

energy. Finally, we averaged the obtained solvation-included energies of all the samples in each IMS (more details on the MPA-MD method are provided in Supplementary Note 2).

This method allows us to calculate all the elementary steps in the presence of the liquid phase with good accuracy. Thorough tests on the reaction energetics including barriers were carried out to verify the reliability of our approach in dealing with aqueous systems. Five kinds of aqueous reaction were compared between our approach and the state-of-the-art constrained MD methods (Table 1 and see Methods); it would, however, be too time consuming to use the constrained MD approach for the water/TiO₂(110) systems. As shown in Table 1, in all the cases, our MPA-MD method gives very similar results to those from the constrained MD method. In particular, the comparable results of water dissociation (0.51 versus 0.54 eV) as well as the good accuracy in estimating the point of zero charge (PZC; Supplementary Note 4) of the rutile(110) surface (5.83 versus experimental 5.5–4.8; ref. ³¹) further demonstrate the feasibility of our method in investigating the OER mechanism at the aqueous/TiO₂(110) interface. We should point out that without our approach it is impossible to investigate the OER mechanism in such detail as we did using first-principles calculations.

The OER mechanism. The whole mechanism of the OER can be broadly divided into two parts: the first is the water dissociation and the formation of surface species, and the second is the oxygen–oxygen bond formation, yielding O₂ (that is, O–O coupling among surface species).

Water dissociation. We first calculated the dissociation of H₂O adsorbed on the surface (H₂O_{ad}) with and without the participation of photoholes h⁺. Three possible pathways were examined as shown in Figs. 1b and 2: (i) H₂O_{ad} donates a H to a nearby bridge oxygen O_{br}²⁻ (black curve in Fig. 2), forming a terminal hydroxyl OH_i[•] and an O_{br}H[•], which is the widely suggested mechanism^{10,32}; (ii) H₂O_{ad}

Table 1 | Comparison of reaction barriers using different methods

Reactions in solutions	E_a^1 (eV) ^a	E_a^2 (eV) ^b
$\text{Pt}(\text{NH}_3)_2\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{Pt}(\text{NH}_3)_2\text{Cl}(\text{H}_2\text{O})^+ + \text{Cl}^-$	0.78	0.75(exp. 0.84–1.07) ⁵⁴
$\text{Na}_n\text{Cl}_n \rightarrow \text{Na}_n\text{Cl}_{n-1}^+ + \text{Cl}^-$	0.15	0.11 ⁵⁵
$\text{O}_2 \rightarrow 2\text{O}^*$ on Pt(111)	0.43	0.39
$\text{O}_2 + \text{H}^* \rightarrow \text{OOH}^*$ on Pt(111)	0.60	0.58
$\text{H}_2\text{O}^* \rightarrow \text{OH}^* + \text{H}^*$ on TiO_2 rutile (110)	0.51	0.54

Reaction barriers are computed using two methods, ^aour MPA-MD method and ^bthe constrained MD method, in the presence of water solvents. E_a , reaction barrier Exp., experimental value.

releases a proton directly into solution (red curve), yielding an OH_i^- and a proton in water $\text{H}^+(\text{sol})$; and (iii) $\text{H}_2\text{O}_{\text{ad}}$ deprotonates (loses a proton) in the presence of h^+ (blue curve), producing an $\cdot\text{OH}_i$ radical and a $\text{H}^+(\text{sol})$.

It was found that in the presence of liquid water (Fig. 2 and Supplementary Table 1), pathway (i) is in fact not favoured, being endothermic by 0.36 eV with a high barrier of 0.97 eV (TS9 in Fig. 2). Instead, pathway (ii) is more favoured; $\text{H}_2\text{O}_{\text{ad}}$ can deprotonate directly into solution through a Grotthuss-type proton transfer³³ with a barrier of 0.51 eV and an enthalpy change (ΔH) of 0.11 eV at neutral pH. At the transition state (TS), the detaching H^+ is bounded by water molecules, forming a Zundel-like (H_5O_2^+) structure (TS1 in Fig. 2). Then, after a series of catch-and-abandon-like proton transfer across the Grotthuss chain, it forms a stable H_5O_2^+ species in the bulk water and leaves an OH_i^- on the surface. Interestingly, in pathway (iii), the participation of h^+ can further lower the deprotonation barrier to 0.39 eV, and the reaction also becomes exothermic ($\Delta H = -0.08$ eV, pH = 7). At the TS, the h^+ is found to be shared by the OH group and a nearby O_{3c} (TS8 in Fig. 2 and Supplementary Fig. 5), which could facilitate the water deprotonation by weakening the $\cdot\text{OH}-\text{H}^+$ bond and stabilizing the OH adsorption structure²⁷. Although pathway (iii) appears to be the most favoured one, the concentration of surface-reaching holes (C_{h^+} ; in monolayer, ML; ignoring the surface recombination) is so low in TiO_2 ($C_{\text{h}^+} \approx 10^{-9}$ ML)³⁴ that the kinetics of this concerted-like water dissociation involving h^+ ($\text{H}_2\text{O} + \text{h}^+ \rightarrow \cdot\text{OH}_i + \text{H}^+(\text{sol})$) is in fact less favoured (Supplementary Note 3). Our kinetic modelling shows that it is about three orders of magnitude slower than the direct deprotonation (that is, pathway (ii)) followed by a hole transfer process, being consistent with the results on anatase TiO_2 (ref. ⁹).

Surface radicals. After identifying the favoured pathway of water dissociation yielding OH_i^- , we investigated the formation of surface radicals (see Methods), for example $\cdot\text{OH}_i$ and O_{br}^- (Fig. 3d), which are the key species in the photocatalytic OER. It was found that the hole trapping at OH_i^- ($\text{OH}_i^- + \text{h}^+ \rightarrow \cdot\text{OH}_i$) is 0.24 eV more favourable than that observed for $\text{O}_{\text{br}}^{2-}$ ($\text{O}_{\text{br}}^{2-} + \text{h}^+ \rightarrow \text{O}_{\text{br}}^-$). Additionally, compared to the situation in the gas phase²⁷, it is worth noting that the absolute values of the hole-trapping capacity of surface hole traps (for example OH_i^- and $\text{O}_{\text{br}}^{2-}$) are remarkably reduced by ~ 0.6 eV in the presence of liquid water. This indicates that those surface hole traps are stabilized by the surrounding water so much that they are much less prone to trap h^+ . Furthermore, it is interesting to note that once the $\cdot\text{OH}_i$ radical is formed, it can readily transform into the O_i^- radical on the surface via detaching a H^+ into the solution^{13,35} with a barrier as low as 0.41 eV at neutral pH (TS2 in Fig. 3a and Supplementary Fig. 5). The existence of the O_i^- radical is made evident in our calculated spin charge density (Fig. 3d), and the ease of the O_i^- formation from the $\cdot\text{OH}_i$ radical is in accordance with the experimental results³⁶.

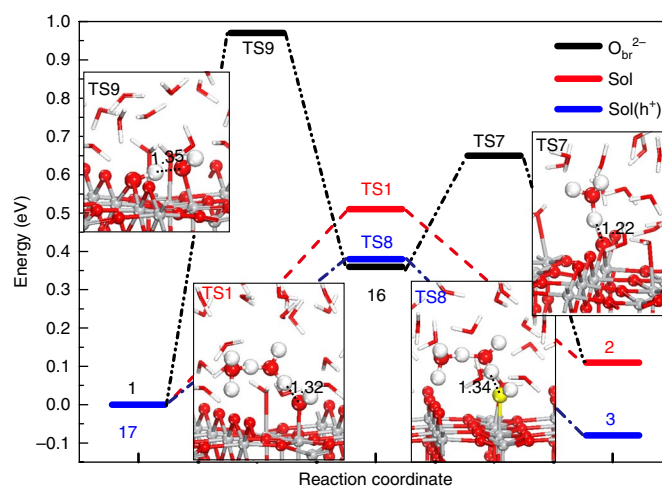


Fig. 2 | Energy profiles of possible water dissociation pathways. Three pathways are considered under aqueous surrounding at the HSE06 level: deprotonation to a nearby $\text{O}_{\text{br}}^{2-}$ first and then into solution (black), deprotonation into solution directly (red), and deprotonation into solution with the help of a hole (blue). The TS structures are shown in the insets with surface radicals highlighted in yellow, and other related structures (IMS 1, 2, 3, 16, 17) are shown in Supplementary Fig. 4. Length unit: Å.

O–O coupling. Regarding the second part of the mechanism, that is, the coupling of the surface species, all the possible couplings were considered in the current work. We found that all the non-radical species (OH_i^- or $\text{O}_{\text{br}}^{2-}$) are inert for coupling with each other, even including one non-radical coupling with one radical, due to the saturation of valence electrons. For example, the coupling barriers (E_a^{coup}) of $\text{O}_i^-/\text{O}_{\text{br}}^{2-}$ (short for coupling between O_i^- and $\text{O}_{\text{br}}^{2-}$), $\cdot\text{OH}_i/\text{O}_{\text{br}}^{2-}$ and $\text{O}_i^-/\text{OH}_i^-$ are all higher than 1.5 eV with large endothermic reaction energies of 2.55, 1.74 and 1.35 eV, respectively. Thus, the only promising O–O couplings reside in the pairing among surface radicals ($\cdot\text{OH}_i$, O_i^- and O_{br}^-), corresponding to a total of six possibilities: (i) $\text{O}_i^-/\text{O}_i^-$; (ii) $\cdot\text{OH}_i/\text{O}_i^-$; (iii) $\cdot\text{OH}_i/\cdot\text{OH}_i$; (iv) $\cdot\text{OH}_i/\text{O}_{\text{br}}^-$; (v) $\text{O}_{\text{br}}^-/\text{O}_{\text{br}}^-$ and (vi) $\text{O}_i^-/\text{O}_{\text{br}}^-$, as shown in Fig. 1c.

Our DFT calculations show that the $\text{O}_{\text{br}}^-/\text{O}_{\text{br}}^-$ pathway is not likely, owing to the infeasible thermodynamics in producing two neighbouring O_{br}^- radicals. Also, neither is the $\text{O}_i^-/\text{O}_{\text{br}}^-$ pathway due to the relatively long distance at the TS between O_i^- and O_{br}^- with a large E_a^{coup} of ~ 1.0 eV. For the $\cdot\text{OH}_i/\cdot\text{OH}_i$ pathway previously proposed⁴, our calculations show that the E_a^{coup} is 0.66 eV, which is higher than the deprotonation barrier of the $\cdot\text{OH}_i$ radical (0.41 eV). In other words, $\cdot\text{OH}_i$ is more inclined to deprotonate into the O_i^- radical rather than to couple with another adjacent $\cdot\text{OH}_i$ radical. The remaining three possible pathways are compared in Fig. 4 and Supplementary Table 2. One can see that the $\cdot\text{OH}_i/\text{O}_i^-$ pathway is not easy because of a high E_a^{coup} of 0.71 eV (TS10 in Fig. 4), while the remaining two pathways ($\text{O}_i^-/\text{O}_i^-$ and $\text{OH}_i/\text{O}_{\text{br}}^-$) are both favoured with the barriers of ~ 0.3 eV (TS3, TS4 in Fig. 4). The necessity of radicals (Supplementary Fig. 5) in the O–O bond formation in our work strongly shows the indispensable role of radicals in the photocatalytic OER. In addition, it is generally more difficult for the O–O coupling to occur at the liquid/solid interface, compared with the case without liquid water where all the barriers are below 0.15 eV (Supplementary Table 2), demonstrating the constraint effect of the H-bond network on the coupling reactions.

The general picture of the OER. By combining all the elementary steps that have been calculated, an OER mechanism with dual pathways can be obtained as shown in Fig. 3: one of the dual pathways occurs on Ti-row sites (the blue curve, named as pathway I) and the other

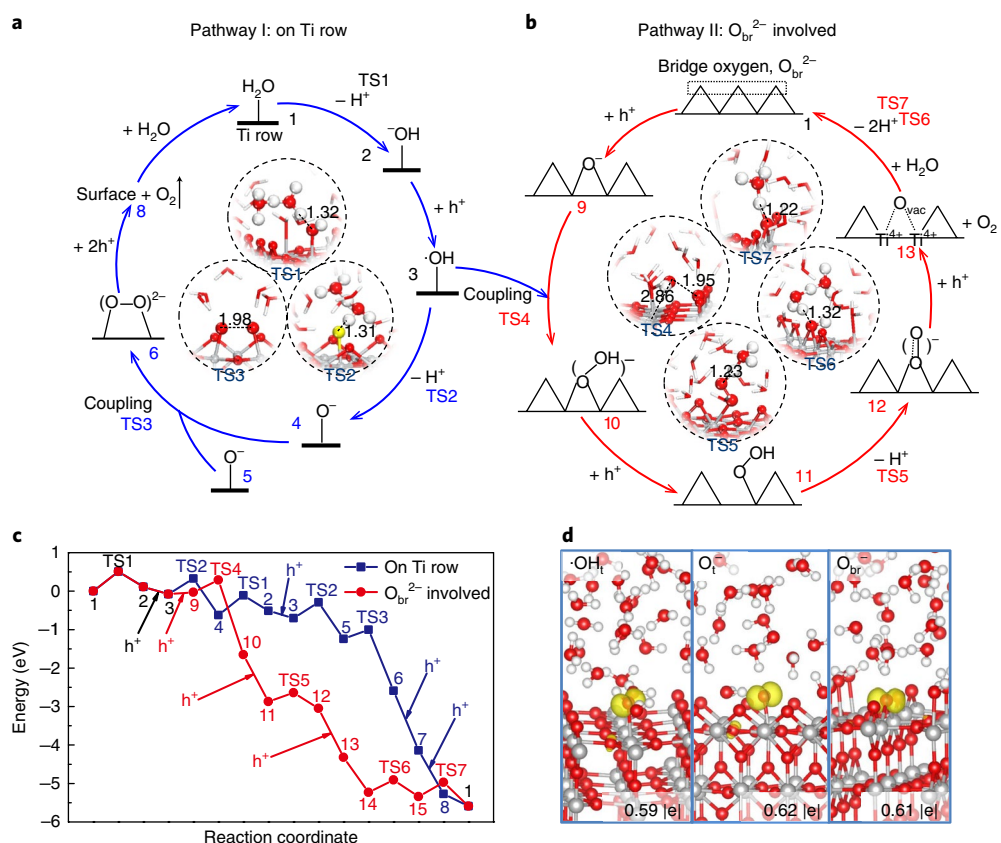


Fig. 3 | Proposed mechanism (dual pathways) and energy profiles for the photocatalytic OER. **a**, One of the dual pathways (pathway I) occurring on the Ti row. **b**, The other pathway (pathway II) involving bridge oxygen. Insets show the TS structures, and all of the detailed information of the structures is shown in Supplementary Figs. 4 and 5. **c**, The energy profiles of pathways I and II, in which states 1, 2, ...15 correspond to the states in **a** and **b**. The elementary steps involving holes are labelled by h^+ . **d**, The existence of three key surface radicals in the process. They are illustrated by the spin-density plots (dumbbell-shaped O 2p orbital; isovalue of 0.005) and the Bader charge difference (~ 0.6 |e| means that those oxygens after trapping h^+ are positively charged with $+0.6$ |e| with respect to the lattice O in bulk TiO_2).

involves bridged oxygen (red curve; pathway II). Pathway I (Fig. 3a) starts from the deprotonation of an adsorbed water to generate an OH_i^- , followed by the trapping of a h^+ to form an $\cdot\text{OH}_i$ on the surface. Then, the $\cdot\text{OH}_i$ deprotonates readily, yielding a O_i^- radical. After coupling with another O_i^- nearby, two successive h^+ can oxidize the as-formed $(\text{O}_i-\text{O}_i)^{2-}$ into an O_2 . On the other hand, in pathway II (Fig. 3b), the $\cdot\text{OH}_i$ radical may couple with an $\text{O}_{\text{br}}^{2-}$ directly instead of deprotonation, and thus the lattice oxygen is involved in the OER. Then, after a sequence of h^+ trapping and deprotonation, the O_2 molecule is generated and readily released, leaving an ionized oxygen vacancy ($\text{O}_{\text{vac}}^{2+}$) on the surface. The generated $\text{O}_{\text{vac}}^{2+}$ can be easily filled by the adsorption of H_2O with two sequential deprotonation steps, recovering the surface. The following points should be noted: (i) pathway II involves lattice oxygen and thus it may cause surface roughening, which will be discussed later; (ii) the requirement of two adjacent radicals for O–O coupling and the successive oxidation of O_2^{2-} into O_2 in the OER mechanism indicate that the surface needs to accumulate some holes to effectively drive the multi-hole process, consistent with experimental results from rate law analysis^{37,38}. In general, the sequential oxidation of O_2^{n-} species by h^+ ($\text{O}_2^{2-} \rightarrow \text{O}_2^- \rightarrow \text{O}_2$; Supplementary Fig. 6) is clearly accompanied by the step-by-step decrease in the bond length of O–O ($l_{\text{O-O}}$: 1.48 $\rightarrow$ 1.35 $\rightarrow$ 1.23 Å) as well as the progressively stretched $l_{\text{Ti-O}}$ (ref. ³⁹), as shown in Table 2, thus indicating the gradual detachment of an oxygen from the surface.

Kinetic analysis. Figure 3c shows the energy profiles of these two pathways and it may seem that both pathways should be kinetically

fast because of the low barriers (the highest barrier is 0.51 eV) and large energy releases (Supplementary Notes 5 and 6). This is in contrast with the general consensus that the performance of TiO_2 is still very low for the photocatalytic OER. How can we understand this? What is the main obstacle that limits the overall efficiency of the OER on TiO_2 ? Having obtained the complete pathways and the energetics of the elementary steps in the OER, we were able to carry out the kinetic analysis utilizing microkinetics to address these issues quantitatively⁴⁰.

We first estimated the reaction kinetics under the experimental concentration of surface-reaching holes ($C_{h^+} = \sim 10^{-9}$ ML) by utilizing a steady-state microkinetic model within the framework of transition-state theory. All the considered reactions and energies are shown in Table 3, where * represents the free site of Ti_{5c} on the Ti row and $\circ$ represents the free site of O_{vac} on the O_{br} row, corresponding to the elementary steps in Fig. 3c. The total coverage of all intermediate species adsorbed on each kind of surface site (Ti_{5c} on the Ti row or O_{vac} on the O_{br} row) are treated as unity, respectively. The kinetic equations are solved under the condition of $\chi_{\text{H}_2\text{O}} = 1$ (mole fraction), $\chi_{\text{O}_2} = 10^{-7}$, pH = 7 and $T = 300$ K. Differing from the oxygen reduction reaction ($\chi_{\text{O}_2(\text{aq})} = 2.34 \times 10^{-5}$ corresponding to 1 atm $\text{O}_{2(\text{g})}$ in equilibrium with $\text{O}_{2(\text{aq})}$)⁴¹, a relatively smaller χ_{O_2} value is used because of the extremely unsaturated situation of O_2 in water caused by the slow reaction kinetics of the OER (Supplementary Note 7). More importantly, we further investigated the influences of different kinetic barriers (that is, water dissociation, O–O coupling and hole diffusion) and/or C_{h^+} on the total turnover frequency (TOF,

Table 2 | Geometric properties of key species involved in the dual pathways

Item	$N_{\text{Ti-O}}^a$	$l_{\text{Ti-O}}^b$	$l_{\text{O-O}}^c$	$\nu_{\text{O-O}}^d$	Item	$N_{\text{Ti-O}}^a$	$l_{\text{Ti-O}}^b$	$l_{\text{O-O}}^c$	$\nu_{\text{O-O}}^d$
OH_i^-	1	1.83	–	–	$\text{O}_{\text{br}}^{2-}$	2	1.85;1.85	–	–
$\cdot\text{OH}_i$	1	2.25	–	–	O_{br}^-	2	2.06;2.06	–	–
O_i^-	1	2.00	–	–	$(\text{O}_{\text{br}}-\text{OH})^\cdot$	2	2.11;2.14	1.48	816
$(\text{O}_i-\text{O}_i)^{2-}$	2	1.97;2.00	1.46	840	$(\text{O}_{\text{br}}-\text{OH})$	1	2.37	1.34	1145
$(\text{O}_i-\text{O}_i)^\cdot$	1	2.25	1.34	1108	$(\text{O}-\text{O}_{\text{br}})^\cdot$	2	2.26;2.38	1.35	1115
$(\text{O}_i-\text{OH})^\cdot$	1	2.00	1.47	928	$\text{O}_2(\text{aq})$	0	–	1.23	1492

All the results are calculated in the presence of aqueous solution, including: ^anumber of Ti–O bonds; ^bbond length of Ti–O bonds and ^cbond length of O–O bonds (in Å); and ^dstretching frequency (in wavenumber: cm^{-1}) of O–O bonds.

which is a quantitative measure of the overall rate), by applying a similar approach with the method proposed in ref. ⁴² (that is, degree of rate control). For instance, when investigating the influence of the water dissociation barrier and/or C_{h+} on the total TOF (Fig. 5a), we keep the barriers of O–O coupling and hole diffusion constant, meanwhile mapping the TOF value (solve the kinetic equations at every single point) in the vertical range of (0, 1.5) for the dissociation barrier at intervals of 0.01 and in the horizontal range of (–10, 0) for the $\log_{10} C_{h+}$ at intervals of 0.1, corresponding to a mesh density of 150×100 in the figure.

Our kinetic analysis results of investigating the influence of different kinetic barriers and C_{h+} on the TOF value are shown in Fig. 5. They show that there are different regions in which the TOF is affected by (i) only C_{h+} (region I); (ii) only the barrier (region II); (iii) neither C_{h+} nor the barrier (region III); and (iv) both C_{h+} and the barrier (region IV). Several striking features are revealed. First, at the experimental concentration of surface-reaching holes ($C_{h+} \sim 10^{-9}$ ML)³⁴, the total TOF would be dramatically hindered, being five orders of magnitude lower compared with the case when photoholes are always available ($C_{h+} = 1$ ML) as shown in Fig. 5d. This explains the origin of low activity observed experimentally. Second, there is a considerable margin where all the reaction barriers on TiO_2 can be increased without reducing the TOF considerably. For example, the water dissociation barrier can increase from 0.51 eV (the DFT value on TiO_2) to ~ 0.8 eV without affect-

ing the TOF at $C_{h+} = 10^{-9}$. Similarly, the TOF is roughly constant if the O–O coupling barrier is below 0.8 eV (the DFT value is only ~ 0.3 eV). This is a surprising finding, which provides strong theoretical evidence to show that under experimental conditions the inherent catalytic activity of TiO_2 is adequate (that is, the barriers are low enough) and the main obstacle that limits the overall efficiency of the OER is in fact the low concentration of surface-reaching holes. Third, lowering the barrier of water dissociation can increase the TOF once C_{h+} is higher than $\sim 10^{-4}$, shown in Fig. 5a. In other words, to further increase the TOF one must increase the catalytic activity of TiO_2 (lower the barriers), if C_{h+} is higher than $\sim 10^{-4}$. This suggests that $C_{h+} \sim 10^{-4}$ is a critical threshold to signify whether the catalytic activity or the concentration of holes needs to be improved.

Our kinetic results also reveal that both the hole diffusion barrier (Supplementary Note 8) and C_{h+} can influence the TOF in region IV (Fig. 5c), and the diffusion barrier threshold between region I and IV is ~ 0.19 eV. Considering that the diffusion barrier in rutile TiO_2 is already so low⁴³, it would be difficult to decrease it further. The most effective way to enhance the photocatalytic efficiency is to increase the concentration of photoholes. Another finding of the kinetic analysis is that the two pathways possess different characteristics of hole-concentration dependence: the pathway involving lattice oxygen (pathway II) tends to reduce its contribution to the total TOF as

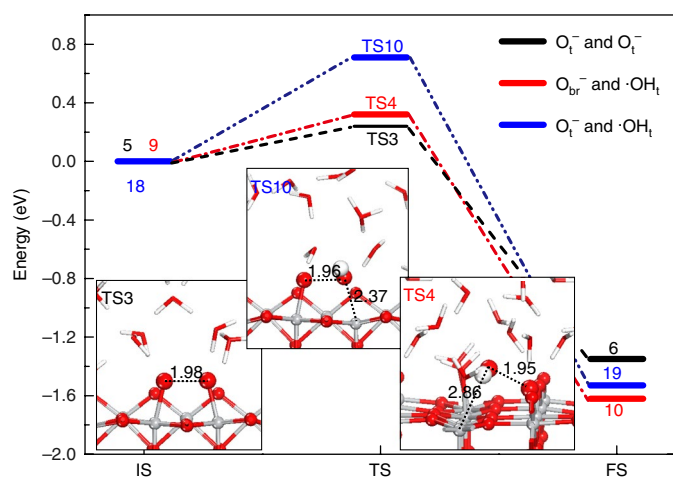


Fig. 4 | Energy profiles of O–O coupling reactions among various surface radicals. Three pathways are considered under aqueous surrounding at the HSE06 level: coupling between $\cdot\text{OH}_i$ and O_{br}^- (red), $\cdot\text{OH}_i$ and O_i^- (blue), and two O_i^- radicals (black). The TS structures are shown in the insets and other related structures (IMS 5, 6, 9, 10, 18, 19) including the initial (IS) and final states (FS) in the O–O coupling are shown in Supplementary Fig. 4. Length unit: Å.

Table 3 | Reaction energetics of each elementary step for solving the kinetic modelling

Reaction equations	Steps	ΔH (eV)	E_a (eV)
$\text{H}_2\text{O}(\text{sol}) + ^\cdot \rightarrow ^\cdot\text{OH}^- + \text{H}^+(\text{sol})$	1 $\rightarrow$ 2	0.11	0.51
$^\cdot\text{OH}^- + \text{h}^+ \rightarrow ^\cdot\text{OH-rad}$	2 + $\text{h}^+ \rightarrow$ 3	–0.19	– ^a
$^\cdot\text{OH-rad} \rightarrow ^\cdot\text{O}^- + \text{H}^+(\text{sol})$	3 $\rightarrow$ 4	–0.54	0.41
$^\cdot\text{O}^- + ^\cdot\text{O}^- \rightarrow ^\cdot\text{O}_2^{2-} + ^\cdot$	5 $\rightarrow$ 6	–1.35	0.24
$^\cdot\text{O}_2^{2-} + \text{h}^+ \rightarrow ^\cdot\text{O}_2^-$	6 + $\text{h}^+ \rightarrow$ 7	–1.55	– ^a
$^\cdot\text{O}_2^- + \text{h}^+ \rightarrow \text{O}_2(\text{aq}) + ^\cdot$	7 + $\text{h}^+ \rightarrow$ 1	–1.45	– ^a
$^\cdot\text{O}_{\text{br}}^{2-} + \text{h}^+ \rightarrow ^\cdot\text{O}_{\text{br}}^-$	3 + $\text{h}^+ \rightarrow$ 9	0.05	– ^a
$^\cdot\text{O}_{\text{br}}^- + ^\cdot\text{OH-rad} \rightarrow ^\cdot\text{O}_{\text{br}}\text{OH}^\cdot + ^\cdot$	9 $\rightarrow$ 10	–1.62	0.32
$^\cdot\text{O}_{\text{br}}\text{OH}^\cdot + \text{h}^+ \rightarrow ^\cdot\text{O}_{\text{br}}\text{OH}$	10 + $\text{h}^+ \rightarrow$ 11	–1.22	– ^a
$^\cdot\text{O}_{\text{br}}\text{OH} \rightarrow ^\cdot\text{O}_{\text{br}}\text{O}^- + \text{H}^+(\text{sol})$	11 $\rightarrow$ 12	–0.18	0.23
$^\cdot\text{O}_{\text{br}}\text{O}^- + \text{h}^+ \rightarrow \text{O}_2(\text{aq}) + ^\cdot$	12 + $\text{h}^+ \rightarrow$ 13	–1.27	– ^a
$\text{H}_2\text{O}(\text{sol}) + ^\cdot \rightarrow ^\cdot\text{H}_2\text{O}$	13 $\rightarrow$ 14	–0.91	– ^a
$^\cdot\text{H}_2\text{O} \rightarrow ^\cdot\text{OH}^- + \text{H}^+(\text{sol})$	14 $\rightarrow$ 15	–0.11	0.32
$^\cdot\text{OH}^- \rightarrow ^\cdot\text{O}_{\text{br}}^{2-} + \text{H}^+(\text{sol})$	15 $\rightarrow$ 1	–0.25	0.37

All the results are calculated using our MPA-MD method except for the hole diffusion barrier.

^aTaken from ref. ⁴⁶; Supplementary Note 8. Rate equations are shown in Supplementary Table 3.

^{*} represents the free site of Ti_{5c} on the Ti row and [#] represents the free site of O_{vac} on the O_{br} row.

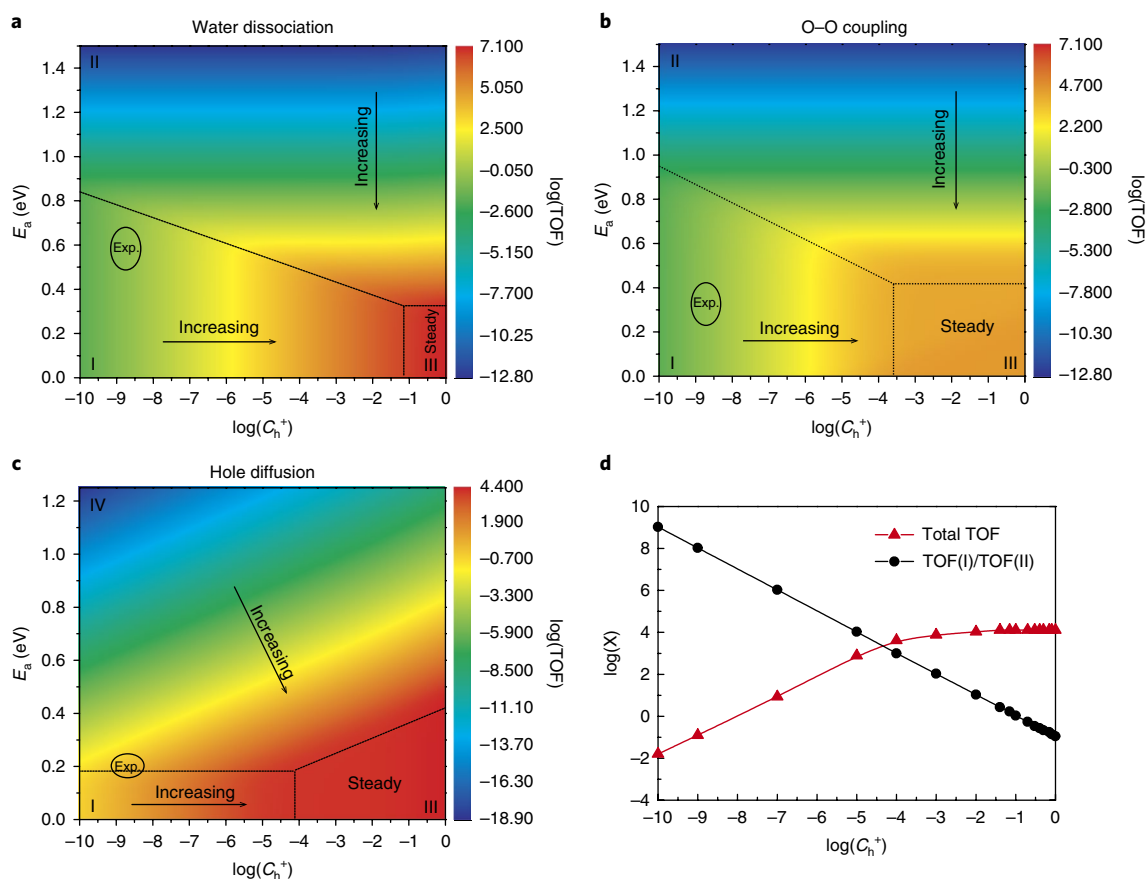


Fig. 5 | TOF as a function of the hole concentration from microkinetic analyses. a–c, Influence of different kinetic barriers (water dissociation (a), O–O coupling (b) and hole diffusion (c)) and C_{h+} on the total TOF (Supplementary Notes 7 and 8). The trends in different regions (I, II, III and IV) are indicated by black arrows. The experimental concentrations of holes with calculated barriers are labelled as black circles. **d,** The contribution ratio between pathways I and II to the total TOF. ‘X’ represents the total TOF for the red curve and the contribution ratio for the black curve, respectively.

C_{h+} is decreased, from the dominant role at $C_{h+}=1$ (~89%) to a negligible one (~ 10^{-8}) at $C_{h+}=10^{-9}$ (Fig. 5d and Supplementary Tables 4 and 5). This change in reaction channels/mechanism varying with C_{h+} was also found in other materials, such as hematite³⁸. It may be worth noting that, despite remarkable progress made previously pointing out that the photoactivity is significantly affected by the hole concentration^{37,38,44}, the analysis above describes quantitatively, by virtue of microkinetic analysis, the effects of each of the kinetic components on the overall efficiency and it may be extremely difficult to obtain these quantities using other approaches.

Insights into the diverse experimental observations. Having performed the kinetic analysis, we are now at the position to rationalize some classical but puzzling experimental observations. First, it was detected that on rutile TiO_2 there are three characteristic multiple internal reflection infrared radiation (MIR-IR) signals⁵: 838 cm^{-1} , 812 cm^{-1} and 928 cm^{-1} . More interestingly, a change of peak intensity under different pH was observed: when switching from acid (pH 2.4) to alkaline (pH 11.9) solutions, the peak of 812 cm^{-1} increases while the peak of 928 cm^{-1} decreases. However, what surface species are responsible for these signals and why the two peak intensities shift are still elusive. The current work can shed some light on these issues: our simulated stretching frequencies of key O–O species (see Methods) are listed in Table 2, which shows that the three frequencies of 840 cm^{-1} , 816 cm^{-1} and 928 cm^{-1} agree well with the experimental signals, identifying the corresponding species (that is, $(\text{O}_t-\text{O}_l)^{2-}$, $(\text{O}_{br}-\text{OH})^-$ and $(\text{O}_t-\text{OH})^-$, respectively) for these signals. Besides, the peak intensity changes can be

understood by taking the pH dependence of the concentrations of surface radicals into account. For instance, the peak growth of 812 cm^{-1} (corresponding to $(\text{O}_{br}-\text{OH})^-$), is attributed to the rise of O_{br}^- concentration as pH increases: increasing the pH from 2.4 to 11.9 would increase the concentration of O_{br}^{2-} that is easily oxidized by h^+ to form O_{br}^- , and further promote the $\text{O}_{br}^-/\text{OH}_t$ coupling, leading to an increase of $(\text{O}_{br}-\text{OH})^-$. The peak intensity change of 928 cm^{-1} can be explained in a similar way.

Second, we can rationalize the well-known observation of a sharp decrease of photoluminescence (PL) intensity⁸ around pH 4: the availability of surface O_{br}^{2-} increases as pH goes above 4.3 (ref. ⁴⁵), thus facilitating the pathway involving the lattice oxygen and greatly suppressing the recombination of photoinduced carriers. As pH is further increased, all the deprotonation steps are promoted, corresponding to the further decrease of PL intensity, which even approaches zero near pH 13. Third, another experimental observation whereby the surface roughening on rutile(110) was significantly suppressed at pH 13 compared with pH 1.1 (ref. ⁸) can also be understood by our findings: the O_{br}^- radical can only couple with $\cdot\text{OH}_t$ instead of O_t^- , and thus the lack of $\cdot\text{OH}_t$ in a strong alkaline solution reduces the possibility of $\text{O}_{br}^-/\cdot\text{OH}_t$ coupling, leading to the decrease in the rate of pathway II and hence a reduction of surface roughening. Similarly, our result showing that pathway II is favoured if C_{h+} is high can explain the experimental fact that low-intensity UV illumination causes little surface roughening on rutile(110), and it becomes more prominent with the increasing illumination intensity (from 0.04 to 50 mW cm^{-2})⁷. All these results strongly support our proposed mechanism.

Discussion

It is worth discussing some implications of our results in photocatalytic water splitting. In the photocatalytic OER, the radicals are perhaps the most important intermediates; however, in which manner they play this essential role has remained elusive. This fundamental question can be answered as follows. First, radicals can reduce the Ti–O bonding strength in accordance with the stretched $l_{\text{Ti-O}}$ of radicals as shown in Table 2, and thus could facilitate the desorption of O_2 . For instance, the oxygen vacancy formation energy of $\text{O}_{\text{br}}^{2-}$ (the normal bridge oxygen) and $\text{O}_{\text{br}}^{\cdot}$ (the bridge oxygen radical) were calculated to be 3.16 eV and 1.54 eV, respectively, which clearly reveals the contributions of radicals in promoting the removal of lattice oxygen. Second, they are irreplaceable intermediates for O–O coupling where non-radical species are inert. In particular, it is worth mentioning the specific role of the $\cdot\text{OH}_i$ radical in the process. As mentioned before, the favoured O–O coupling on the Ti row is between two $\text{O}_i^{\cdot}$ radicals, and hence the difficulty in producing $\text{O}_i^{\cdot}$ largely decides the feasibility of this pathway. However, the direct deprotonation of $\text{OH}_i^{\cdot}$ can hardly occur due to the fast reverse reaction with a very low barrier of 0.08 eV. Therefore, the only way is to trap a h^+ and form $\cdot\text{OH}_i$ first, followed by the deprotonation of $\cdot\text{OH}_i$ into the $\text{O}_i^{\cdot}$ radical. In other words, the ease of deprotonation of $\cdot\text{OH}_i$ influences the main pathway of the OER under experimental conditions. Third, radicals are actually trapped holes on the surface, which can significantly suppress the surface electron–hole recombination and increase the lifetime of surface-reaching holes up to microseconds^{46,47}.

Here, we developed a first-principles MPA-MD method to extensively simulate the OER in the presence of liquid water on rutile $\text{TiO}_2(110)$. A complete reaction mechanism with dual pathways is identified in this work, which allowed us to join some detached experimental observations and to rationalize the experimental results. Moreover, we provide strong theoretical evidence to show that the low efficiency of the OER on $\text{TiO}_2(110)$ is in fact not limited by the inherent catalytic activity (reaction barriers), but by the low concentration of photoholes on TiO_2 . Perhaps the low concentration of surface-reaching photoholes may be a general obstacle in many other materials⁴⁸. In addition, this finding also explains that rutile TiO_2 is a very good coating material for the photoanode in protecting the effective but unstable light-harvesting semiconductors from corrosion by virtue of its high kinetic activity, stable structure and transparent property⁴⁸. It is clear that the most efficient way to improve the performance of the OER at the current stage (before approaching the estimated threshold of $C_{\text{h}^+} \sim 10^{-4}$) is to enrich the concentration of surface-reaching photoholes rather than to lower the reaction barriers of water dissociation and O–O coupling. The following two strategies identified in the current work may help realize this goal in addition to conventional methods; to lower the hole diffusion barrier and to reduce the hole diffusion distance, that is, achieving surface or near-surface photoexcitation.

Methods

DFT calculations. All the spin-polarized calculations were performed with the Perdew–Burke–Ernzerhof (PBE) functional using the VASP code^{49,50}. The project-augmented wave (PAW) method was used to represent the core–valence electron interaction with electrons from Ti 3p, 3d, 4s; O 2s, 2p; and H 1s shells explicitly included. The valence electronic states were expanded in plane wave basis sets with energy cutoff of 450 eV, and the occupancy of the one-electron states was calculated using the Gaussian smearing with SIGMA = 0.05 eV. The ionic degrees of freedom were relaxed using the BFGS minimization scheme until the Hellman–Feynman forces on each ion were less than 0.05 eV Å^{−1}. The transition states were searched using a constrained optimization scheme, and were verified when (i) all forces on atoms vanish; and (ii) the total energy is a maximum along the reaction coordinate but a minimum with respect to the rest of the degrees of freedom. The force threshold for the TS search was 0.05 eV Å^{−1}. The dipole correction was performed throughout the calculations to take the polarization effect into account.

Regarding the calculation on the stretching frequency of O–O bonds: first, we fully optimized the structures (with water solution included) until the

force on each atom is less than 0.01 eV Å^{−1}. Then, we fixed all the atoms except the relevant O atoms, and calculated the stretching frequency of each O–O bond by setting IBRION = 5. It should be noted that in calculating the frequency, a very small movement (POTIM = 0.001) but a very high convergence criteria (EDIFFG = 1×10^{-6}) are necessarily applied. Finally, six frequencies located in different wavenumber ranges for each structure are obtained, and only the frequency values in the experimental detection range are listed in Table 2.

MD simulations. The MD calculation was performed using a four-Ti-layer $p(1 \times 4)$ periodical rutile(110) slab with a ~ 15 Å vacuum between slabs, and a corresponding $1 \times 2 \times 1$ K-points mesh was used during optimizations. A lattice-matching bulk ice (containing 26 H_2O) was introduced above the surface as an initial aqueous network (with a density similar with liquid water) at the liquid/solid interface. The simulation temperature was set at 300 K (experimental temperature) with a 0.5 fs movement for each step in the canonical (NVT) ensemble employing Nosé–Hoover thermostats. For the MD simulation of each IMS (~ 9 ps), we found that the water structures in our systems often change significantly in the first ~ 2 ps, and reach quasi-equilibrium around 3 ps, as indicated by the black arrow in Supplementary Fig. 2, taking the MD simulation for the water/ $\text{TiO}_2(110)$ interface structure as an example. Then we ran the MD simulation for a period of ~ 6 ps to confirm that the systems were in equilibrium. The structures of the input ice-like water and the water structure after the MD simulation are shown as the insets. One can see that the initial ice-like structure has a perfectly repeated six-member ring structure in the water layer, but lacks a close interaction with the TiO_2 surface. However, the water layer after the MD simulation is relatively less structured with distorted six-, five-, or four-member ring structures, but the water/ TiO_2 interface is well constructed with an enriched density of H atoms as demonstrated by the radial distribution graph in Fig. 1a.

Constrained MD. The constrained MD method was well established on the basis of thermodynamic integration of the free-energy gradient^{51–53}. It is a well-accepted approach to calculate the free-energy change (including reaction barriers). Taking the O_2 dissociation reaction on Pt(111) as an example (reaction 3 in Table 1; Supplementary Fig. 3), the O–O bond distance is stretched gradually from 1.4 to 2.1 Å. For each point per 0.1 Å, we performed ab initio MD simulations at a constant temperature ($T = 300$ K) until the interatomic force between the two constrained atoms (that is, O–O) converged. All the interatomic forces along the reaction coordinate, which are the corresponding free-energy gradients, can be readily obtained. Then by integrating the free-energy gradients, the free-energy change can be computed. In Supplementary Fig. 3, one can see that the highest point in the free-energy profile is located at the O–O distance of 2.0 Å, corresponding to the transition state of O_2 dissociation with the barrier of 0.39 eV.

Localization of photoholes. The localization of a hole on a particular O site of TiO_2 can be obtained by structure optimizations using the BFGS optimization method. Initial magnetic moments on each atom are usually necessary in the input setting, although they will be optimized during the calculation. To ensure the electro-neutrality of the system and to eliminate the possible incorrect energy result from the background charge, trapped holes or surface radicals were simulated by introducing an OH group on the opposite surface instead of extracting electrons out of system in this study²⁷. Besides, the on-site Hubbard U term (DFT+U) was added on O 2p orbitals at the value of 6.3 eV (usually ranging from 3.0 to 7.0 eV) throughout all the MD simulations to hold the spin-polarized property of trapped holes or surface radicals. We further performed HSE06 corrections on all the selected samples from each MD simulation with the electronic minimization algorithm specified to the Damped method and a very soft augmentation charge (PRECFOCK = Fast).

The radical species possess distinctive features relative to their non-radical charged species, and we can distinguish them easily from the following. (i) Magnetic signals: a pure TiO_2 slab has no spin-polarizing features ($\text{mag} = 0$) and the generation of radical species always introduces obvious magnetic signals in the spin-polarized calculation output file ($\text{mag} = 1$). (ii) Geometry structures: the hole localization (generation of radicals) is accompanied by the distinct elongations of Ti–O bonds by the outward movement of the lattice Ti^{4+} ions. We have discussed this aspect in our previous work systematically²⁷. (iii) Electronic structure analysis: the localization of the hole can be further confirmed by the electronic structure analysis of site-projected magnetic moments ($\sim 0.8 \mu_B$) and the Bader charge difference ($\sim 0.6 |e|$; positively charged with $+0.6 |e|$ compared to the lattice O in bulk TiO_2). We have performed electronic analysis on all the hole-related calculations as visualized by the spin-density plots (Supplementary Fig. 5).

Data availability. All data are available within the article (and its Supplementary Information files) and from the corresponding authors upon reasonable request.

Code availability. The kinetic code that is used to estimate the catalytic activity and other relevant data within the article is available from the corresponding author or the website <https://github.com/JianFuChen/Kinetic-solver.git>.

Received: 15 September 2017; Accepted: 13 March 2018;
Published online: 16 April 2018

References

- Chen, X., Shen, S., Guo, L. & Mao, S. S. Semiconductor-based photocatalytic hydrogen generation. *Chem. Rev.* **110**, 6503–6570 (2010).
- Thompson, T. L. & Yates, J. T. Jr Surface science studies of the photoactivation of TiO_2 -new photochemical processes. *Chem. Rev.* **106**, 4428–4453 (2006).
- Nozik, A. J. Photoelectrolysis of water using semiconducting TiO_2 crystals. *Nature* **257**, 383–386 (1975).
- Salvador, P. & Decker, F. On the generation of H_2O_2 during water photoelectrolysis at n- TiO_2 . *J. Phys. Chem.* **88**, 6116–6120 (1984).
- Nakamura, R. & Nakato, Y. Primary intermediates of oxygen photoevolution reaction on TiO_2 (rutile) particles, revealed by in situ FTIR absorption and photoluminescence measurements. *J. Am. Chem. Soc.* **126**, 1290–1298 (2004).
- Cai, R. X., Kubota, Y. & Fujishima, A. Effect of copper ions on the formation of hydrogen peroxide from photocatalytic titanium dioxide particles. *J. Catal.* **219**, 214–218 (2003).
- Nakamura, R., Okamura, T., Ohashi, N., Imanishi, A. & Nakato, Y. Molecular mechanisms of photoinduced oxygen evolution, PL emission, and surface roughening at atomically smooth (110) and (100) n- TiO_2 (rutile) surfaces in aqueous acidic solutions. *J. Am. Chem. Soc.* **127**, 12975–12983 (2005).
- Imanishi, A., Okamura, T., Ohashi, N., Nakamura, R. & Nakato, Y. Mechanism of water photooxidation reaction at atomically flat TiO_2 (rutile) (110) and (100) surfaces: dependence on solution pH. *J. Am. Chem. Soc.* **129**, 11569–11578 (2007).
- Chen, J., Li, Y. F., Sit, P. & Selloni, A. Chemical dynamics of the first proton-coupled electron transfer of water oxidation on TiO_2 anatase. *J. Am. Chem. Soc.* **135**, 18774–18777 (2013).
- Zhao, W. N. & Liu, Z. P. Mechanism and active site of photocatalytic water splitting on titania in aqueous surroundings. *Chem. Sci.* **5**, 2256–2264 (2014).
- Li, Y. F., Liu, Z. P., Liu, L. & Gao, W. Mechanism and activity of photocatalytic oxygen evolution on titania anatase in aqueous surroundings. *J. Am. Chem. Soc.* **132**, 13008–13015 (2010).
- Valdés, A., Qu, Z. W., Kroes, G. J., Rossmeisl, J. & Nørskov, J. K. Oxidation and photo-oxidation of water on TiO_2 surface. *J. Phys. Chem. C* **112**, 9872–9879 (2008).
- Cheng, J., Liu, X., Kattirtzi, J. A., VandeVondele, J. & Sprik, M. Aligning electronic and protonic energy levels of proton-coupled electron transfer in water oxidation on aqueous TiO_2 . *Angew. Chem. Int. Ed.* **53**, 12046–12050 (2014).
- Cho, I. S. et al. Codoping titanium dioxide nanowires with tungsten and carbon for enhanced photoelectrochemical performance. *Nat. Commun.* **4**, 1723 (2013).
- Gai, Y., Li, J., Li, S. S., Xia, J. B. & Wei, S. H. Design of narrow-gap TiO_2 : a passivated codoping approach for enhanced photoelectrochemical activity. *Phys. Rev. Lett.* **102**, 036402 (2009).
- Asahi, R., Morikawa, T., Ohwaki, T., Aoki, K. & Taga, Y. Visible-light photocatalysis in nitrogen-doped titanium oxides. *Science* **293**, 269–271 (2001).
- Zhang, J., Xu, Q., Feng, Z., Li, M. & Li, C. Importance of the relationship between surface phases and photocatalytic activity of TiO_2 . *Angew. Chem. Int. Ed.* **47**, 1766–1769 (2008).
- Wang, X. et al. Photocatalytic overall water splitting promoted by an α - β phase junction on Ga_2O_3 . *Angew. Chem. Int. Ed.* **51**, 13089–13092 (2012).
- Zhao, W.-N., Zhu, S.-C., Li, Y.-F. & Liu, Z.-P. Three-phase junction for modulating electron-hole migration in anatase–rutile photocatalysts. *Chem. Sci.* **6**, 3483–3494 (2015).
- Schaub, R. et al. Oxygen vacancies as active sites for water dissociation on rutile $\text{TiO}_2(110)$. *Phys. Rev. Lett.* **87**, 266104 (2001).
- Aschauer, U. et al. Influence of subsurface defects on the surface reactivity of TiO_2 : water on anatase (101). *J. Phys. Chem. C* **114**, 1278–1284 (2010).
- Maeda, K. Direct splitting of pure water into hydrogen and oxygen using rutile titania powder as a photocatalyst. *Chem. Commun.* **49**, 8404–8406 (2013).
- Abe, R., Sayama, K. & Sugihara, H. Development of new photocatalytic water splitting into H_2 and O_2 using two different semiconductor photocatalysts and a shuttle redox mediator IO_3^-/I^- . *J. Phys. Chem. B* **109**, 16052–16061 (2005).
- Zhang, H., Chen, G. & Bahnemann, D. W. Photoelectrocatalytic materials for environmental applications. *J. Mater. Chem.* **19**, 5089–5121 (2009).
- Kafizas, A., Carmalt, C. J. & Parkin, I. P. Does a photocatalytic synergy in an anatase–rutile TiO_2 composite thin-film exist? *Chem. Eur. J.* **18**, 13048–13058 (2012).
- Cheng, J., Sulpizi, M., VandeVondele, J. & Sprik, M. Hole localization and thermochemistry of oxidative dehydrogenation of aqueous rutile $\text{TiO}_2(110)$. *ChemCatChem* **4**, 636–640 (2012).
- Wang, D., Wang, H. & Hu, P. Identifying the distinct features of geometric structures for hole trapping to generate radicals on rutile $\text{TiO}_2(110)$ in photooxidation using density functional theory calculations with hybrid functional. *Phys. Chem. Chem. Phys.* **17**, 1549–1555 (2015).
- Wang, D., Liu, Z.-P. & Yang, W. Proton promoted electron transfer in photocatalysis: key step for photocatalytic hydrogen evolution on metal/titania composites. *ACS Catal.* **7**, 2744–2752 (2017).
- Wood, B. C., Schwegler, E., Choi, W. I. & Ogitsu, T. Hydrogen-bond dynamics of water at the interface with InP/GaP(001) and the implications for photoelectrochemistry. *J. Am. Chem. Soc.* **135**, 15774–15783 (2013).
- Wood, B. C., Schwegler, E., Choi, W. I. & Ogitsu, T. Surface chemistry of GaP(001) and InP(001) in contact with water. *J. Phys. Chem. C* **118**, 1062–1070 (2014).
- Bullard, J. W. & Cima, M. J. Orientation dependence of the isoelectric point of TiO_2 (rutile) surfaces. *Langmuir* **22**, 10264–10271 (2006).
- Sun, C. H., Liu, L. M., Selloni, A., Lu, G. Q. & Smith, S. C. Titania–water interactions: a review of theoretical studies. *J. Mater. Chem.* **20**, 10319–10334 (2010).
- Marx, D. Proton transfer 200 years after von Grothuss: insights from ab initio simulations. *ChemPhysChem* **7**, 1848–1870 (2006).
- Thompson, T. L. & Yates, J. T. Monitoring hole trapping in photoexcited $\text{TiO}_2(110)$ using a surface photoreaction. *J. Phys. Chem. B* **109**, 18230–18236 (2005).
- Cheng, J., VandeVondele, J. & Sprik, M. Identifying trapped electronic holes at the aqueous TiO_2 interface. *J. Phys. Chem. C* **118**, 5437–5444 (2014).
- Micic, O. I., Zhang, Y., Cromack, K. R., Trifunac, A. D. & Thurnauer, M. C. Trapped holes on titania colloids studied by electron paramagnetic resonance. *J. Phys. Chem.* **97**, 7277–7283 (1993).
- Kafizas, A. et al. Water oxidation kinetics of accumulated holes on the surface of a TiO_2 photoanode: a rate law analysis. *ACS Catal.* **7**, 4896–4903 (2017).
- Le Formal, F. et al. Rate law analysis of water oxidation on a hematite surface. *J. Am. Chem. Soc.* **137**, 6629–6637 (2015).
- Li, Y. F. & Selloni, A. Theoretical study of interfacial electron transfer from reduced anatase $\text{TiO}_2(101)$ to adsorbed O_2 . *J. Am. Chem. Soc.* **135**, 9195–9199 (2013).
- Wang, D., Jiang, J., Wang, H.-F. & Hu, P. Revealing the volcano-shaped activity trend of triiodide reduction reaction: a DFT study coupled with microkinetic analysis. *ACS Catal.* **6**, 733–741 (2015).
- Hansen, H. A., Viswanathan, V. & Nørskov, J. K. Unifying kinetic and thermodynamic analysis of $2e^-$ and $4e^-$ reduction of oxygen on metal surfaces. *J. Phys. Chem. C* **118**, 6706–6718 (2014).
- Stegelmann, C., Andreasen, A. & Campbell, C. T. Degree of rate control: how much the energies of intermediates and transition states control rates. *J. Am. Chem. Soc.* **131**, 8077–8082 (2009).
- Deskins, N. A. & Dupuis, M. Intrinsic hole migration rates in TiO_2 from density functional theory. *J. Phys. Chem. C* **113**, 346–358 (2009).
- Migani, A. & Blancafort, L. What controls photocatalytic water oxidation on rutile $\text{TiO}_2(110)$ under ultra-high-vacuum conditions? *J. Am. Chem. Soc.* **139**, 11845–11856 (2017).
- Connor, P. A., Dobson, K. D. & McQuillan, A. J. Infrared spectroscopy of the TiO_2 /aqueous solution interface. *Langmuir* **15**, 2402–2408 (1999).
- Tamaki, Y. et al. Dynamics of efficient electron-hole separation in TiO_2 nanoparticles revealed by femtosecond transient absorption spectroscopy under the weak-excitation condition. *Phys. Chem. Chem. Phys.* **9**, 1453–1460 (2007).
- Rothenberger, G., Moser, J., Gratzel, M., Serpone, N. & Sharma, D. K. Charge carrier trapping and recombination dynamics in small semiconductor particles. *J. Am. Chem. Soc.* **107**, 8054–8059 (1985).
- Hu, S. et al. Amorphous TiO_2 coatings stabilize Si, GaAs, and GaP photoanodes for efficient water oxidation. *Science* **344**, 1005–1009 (2014).
- Kresse, G. & Furthmüller, J. Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set. *Comp. Mater. Sci.* **6**, 15–50 (1996).
- Kresse, G. & Hafner, J. Ab initio molecular-dynamics simulation of the liquid-metal–amorphous-semiconductor transition in germanium. *Phys. Rev. B* **49**, 14251–14269 (1994).
- Sprik, M. & Cicotti, G. Free energy from constrained molecular dynamics. *J. Chem. Phys.* **109**, 7737 (1998).
- Carlioni, P., Sprik, M. & Andreoni, W. Key steps of the cis-platin–DNA interaction: density functional theory-based molecular dynamics simulations. *J. Phys. Chem. B* **104**, 823–835 (2000).
- Bucko, T. Ab initio calculations of free-energy reaction barriers. *J. Phys. Condens. Matter* **20**, 064211 (2008).
- Song, T. & Hu, P. Insight into the solvent effect: a density functional theory study of cisplatin hydrolysis. *J. Chem. Phys.* **125**, 091101 (2006).
- Liu, L. M., Laio, A. & Michaelides, A. Initial stages of salt crystal dissolution determined with ab initio molecular dynamics. *Phys. Chem. Chem. Phys.* **13**, 13162–13166 (2011).

Acknowledgements

D.W. thanks the Chinese Scholarship Council for financial support for living abroad and P.H. thanks the Chinese Government for support from the “Thousands Talents” program. This work was financially supported by National Natural Science Foundation of China (21333003, 21421004, 21622305), Young Elite Scientist Sponsorship Program by the China Association for Science and Technology (YESS20150131), The Shanghai ShuGuang project (17SG30), and the Fundamental Research Funds for the Central Universities (WJ616007).

Author contributions

P.H. and H.-F.W. conceived the project and contributed to the design of the calculations and analyses of the data. D.W. carried out most of the calculations and wrote the first draft of the paper. T.S. and D.W. conducted the tests of the MPA-MD method. J.C. wrote

the kinetic code and contributed to the analyses of data. All the authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information is available for this paper at <https://doi.org/10.1038/s41929-018-0055-z>.

Reprints and permissions information is available at www.nature.com/reprints.

Correspondence and requests for materials should be addressed to H.-F.W. or P.H.

Publisher's note: Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.